

Future Interfaces Bridging Embodied AI and Accessibility

A UIST 2026 Workshop Proposal

Hamza El Alaoui
Carnegie Mellon University
Pittsburgh, Pennsylvania, USA

Harish Ram Nambiappan
University of Texas at Arlington
Computer Science
Arlington, Texas, USA

Jeff Bigham
Apple
AI/ML
Cupertino, California, USA
Carnegie Mellon University
HCI/LTI
Pittsburgh, Pennsylvania, USA

Eshed Ohn-Bar
Boston University
ECE, Human-to-Everything Lab
Boston, Massachusetts, USA

Tapomayukh Bhattacharjee
Cornell University
CS, EmPRISE Lab
Ithaca, New York, USA

Stephanie Valencia
University of Maryland
Information
College Park, Maryland, USA

Max Pascher
TU Dortmund University
Inclusive HRI
Dortmund, Germany

Vinitha Ranganeni
Hello Robot
Martinez, California, USA

Atieh Taheri
Carnegie Mellon University
HCI
Pittsburgh, Pennsylvania, USA

Abstract

This workshop proposal calls for a UIST venue at the intersection of embodied AI, accessibility, and assistive robotics. It focuses on the interface layer through which disabled users understand, instruct, correct, personalize, and trust robotic assistance. The workshop will bring together researchers and practitioners across user interfaces, accessibility, human-robot interaction, robotics, rehabilitation, AR/VR, wearables, and participatory design to identify interaction techniques and research agendas that make assistive robots more usable, accessible, controllable, and empowering.

CCS Concepts

• **Human-centered computing** → **Accessibility systems and tools**; *Human computer interaction (HCI)*; • **Computer systems organization** → *Robotics*.

Keywords

accessibility, embodied AI, assistive robotics, human-robot interaction, workshop

1 Motivation and Scope

Assistive robots increasingly promise to support people with disabilities in activities such as mobility, manipulation, feeding, navigation, communication, rehabilitation, and daily living. Yet the success of these systems depends not only on robot autonomy, sensing, or control, but also on the *interfaces* through which people understand, instruct, correct, personalize, and trust them. Many assistive robots remain difficult to use because their interaction models are designed around technical capabilities rather than disabled users' lived needs, preferences, agency, and access constraints.

This workshop asks: **How can UIST-style interaction techniques make assistive robots more usable, accessible, controllable, and empowering for people with disabilities?** We

aim to bring together researchers across user interfaces, accessibility, human-robot interaction, assistive technology, rehabilitation, robotics, AI, AR/VR, wearables, and participatory design. The workshop will focus on interface questions around assistive human-robot interaction, including speech and natural-language control, multimodal input, shared autonomy, adaptive interfaces, wheelchair-based interaction, robot teleoperation, AR/VR robot interfaces, explanation and feedback, personalization, safety, trust, and disabled-user agency.

2 Why UIST?

UIST is an ideal venue because the workshop is centered on *interactive systems*: how people communicate with robots, how robots expose their capabilities and uncertainty, how users intervene when autonomy fails, and how new interface techniques can make robotic assistance more practical in everyday life. Rather than treating assistive robotics as only a robotics problem, we frame it as a user interface and interaction problem. This creates a natural bridge between UIST, ASSETS, CHI, HRI, rehabilitation engineering, and robotics communities.

3 Proposed Format

We propose a highly interactive full-day workshop for approximately 25–50 participants. The format will emphasize discussion, demonstrations, brainstorming, and prototyping rather than a sequence of long talks. Activities may include:

- **Lightning talks** to introduce key challenges in accessible human-robot interaction.
- **Demo and video sessions** showcasing assistive robot interfaces, prototypes, and interaction techniques.
- **Breakout discussions** around user agency, safety, autonomy, personalization, evaluation, and accessibility.

- **Hands-on design/prototyping activity** where participants develop interface concepts for concrete assistive robotics scenarios.
- **Synthesis session** to identify open research questions, collaboration opportunities, and future community-building steps.

Participants will not be required to submit long position papers. Instead, we plan to use a short interest form asking about their background, relevant interests, desired discussion topics, optional demos, and accessibility needs.

4 Preliminary Agenda

- 9:00–9:20** Welcome, goals, and participant introductions.
- 9:20–10:10** Lightning talks: key challenges in robotics, accessibility, and UI design.
- 10:30–11:30** Demo/video session: assistive robot interfaces and prototypes.
- 11:30–12:15** Breakout 1: What makes current robot interfaces inaccessible?
- 1:30–2:15** Panel: autonomy, agency, safety, and disabled-user control.
- 2:15–3:15** Hands-on prototyping: designing interfaces for assistive robot scenarios.
- 3:30–4:15** Breakout 2: future research agenda and community-building.
- 4:15–5:00** Group synthesis, next steps, and possible collaborative outputs.

5 Intended Audience and Outcomes

The intended audience includes UIST researchers, accessibility researchers, roboticists, HRI researchers, disabled researchers and designers, assistive technology practitioners, students, and industry researchers interested in interactive systems for robotic assistance. Expected outcomes include: (1) a shared research agenda for accessible robot interfaces; (2) a collection of design challenges and scenarios; (3) new collaborations across UIST, accessibility, and robotics communities; and (4) a public workshop website collecting resources, demos, participant interests, and discussion outcomes.

6 Organizers

Hamza El Alaoui Carnegie Mellon University.

Harish Ram Nambiappan University of Texas at Arlington, Computer Science.

Jeff Bigham Apple AI/ML and Carnegie Mellon University HCI/LTI.

Eshed Ohn-Bar Boston University ECE, Human-to-Everything Lab.

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